

Q

LV.0 → LV.1 · AI ENGINEER

9 Q-words every Level 0 → 1 AI engineer must know.

QKV, QLoRA, Quantization, Q-Learning & more. **Plain-English** definitions, one swipe each, plus Q-named tools and minds to follow.

↓ UP NEXT • TERMS

The Q stack.

QKV

QLoRA

Quantization

Q-Learning

Query/Prompt Caching

Qualitative Variable

Quantitative Variable

QML

Q-Table

Q-LEARNING · REINFORCEMENT LEARNING

Q-Learning

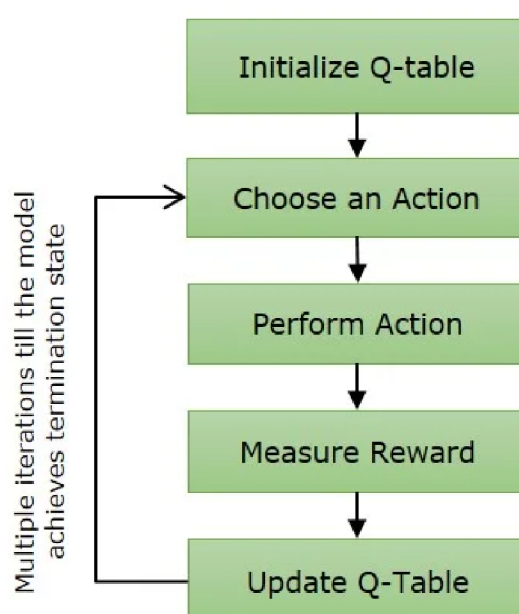
trial, error, reward

Q-Learning teaches an agent to pick optimal actions by **estimating future rewards for every state-action pair**, with no model of the environment required.

USED IN

- Game AI
- Robotics
- RL systems

Q-Learning Algorithm



STATE, ACTION, REWARD LOOP

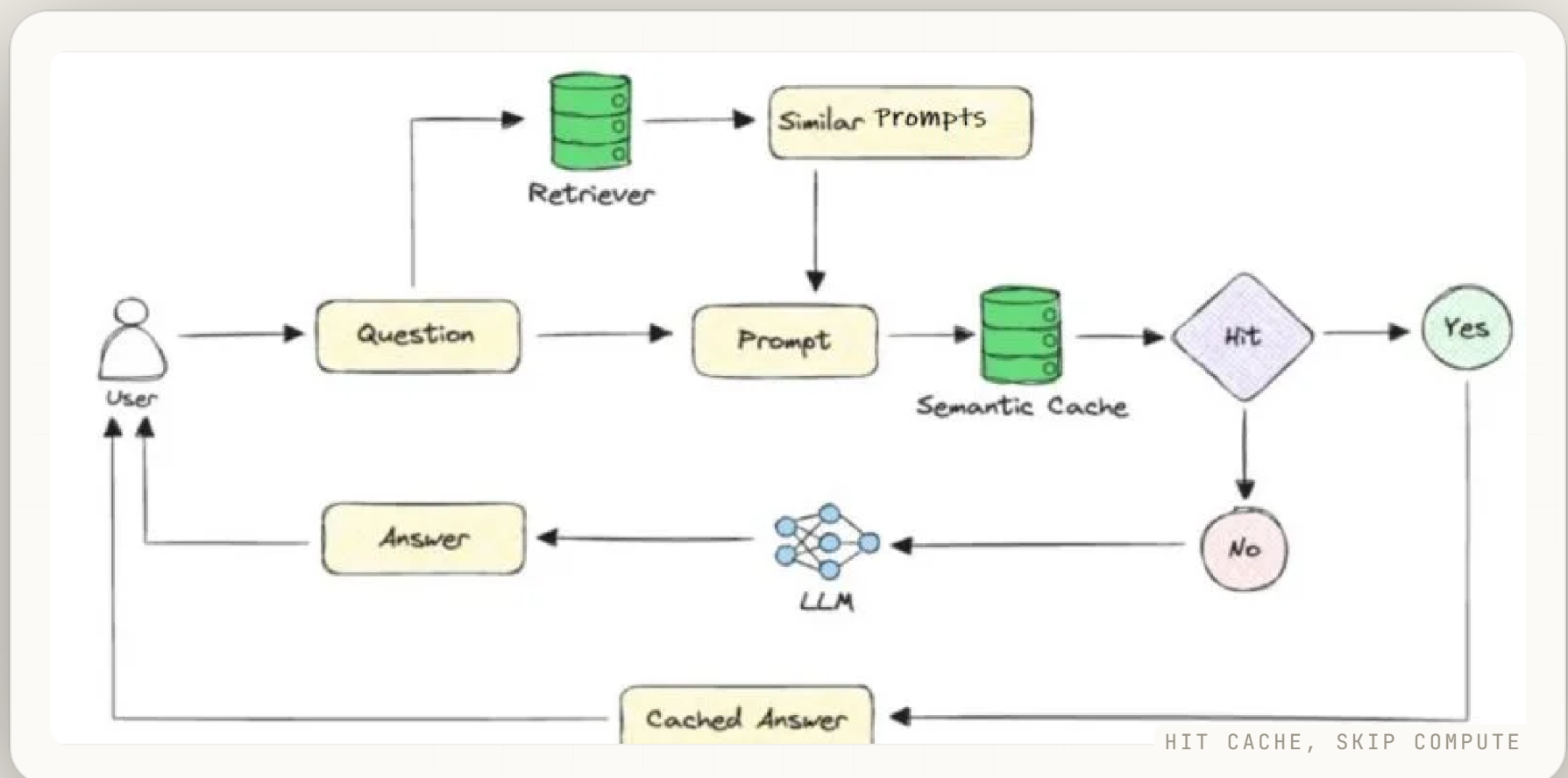
Query / Prompt Caching

compute once, serve forever

Prompt caching stores processed prompt results so identical or repeated inputs **skip recomputation entirely**, cutting latency and inference costs in LLM apps and RAG pipelines.

USED IN

- LLM APIs
- RAG systems
- Cost optimization



QUANTIZED LORA • FINE-TUNING

QLoRA

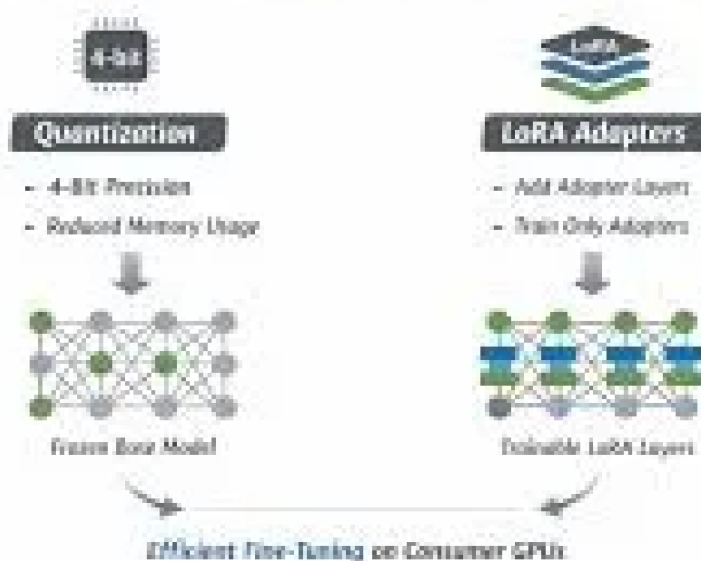
fine-tune big models on small GPUs

QLoRA combines quantization with low-rank adapters so you can **fine-tune a 70B model on a single consumer GPU**, making large-model customization accessible to everyone.

USED IN

- LLM fine-tuning
- Low-resource training
- PEFT

What is QLoRA Training and How Does It Work?



QUANTIZE, THEN ADAPT

QUANTIZATION • OPTIMIZATION

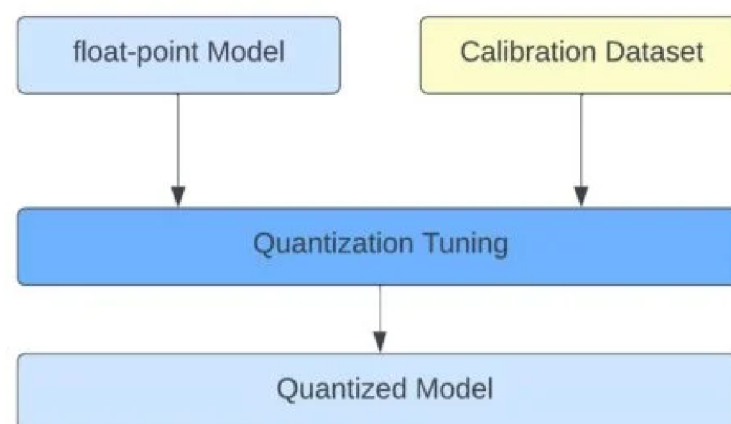
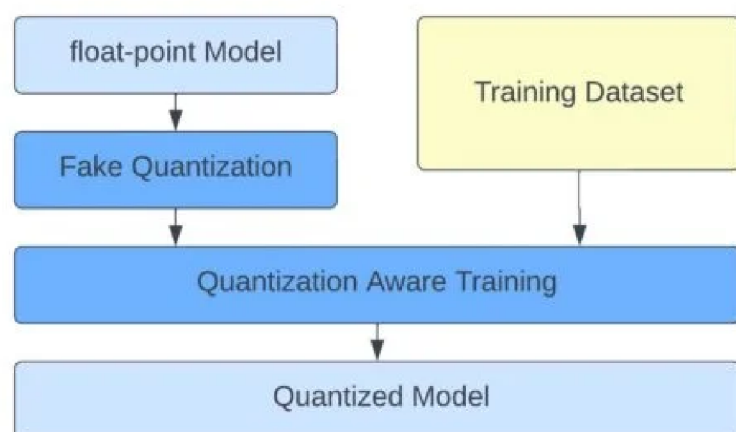
Quantization

fewer bits, faster models

Quantization shrinks model weights from FP32 to INT8 or INT4, **cutting memory and speeding inference** with minimal accuracy loss, making large models viable on edge hardware.

USED IN

- Edge inference
- Model compression
- On-device AI



FP32 TO INT8 TO INT4

QUERY, KEY, VALUE • TRANSFORMERS

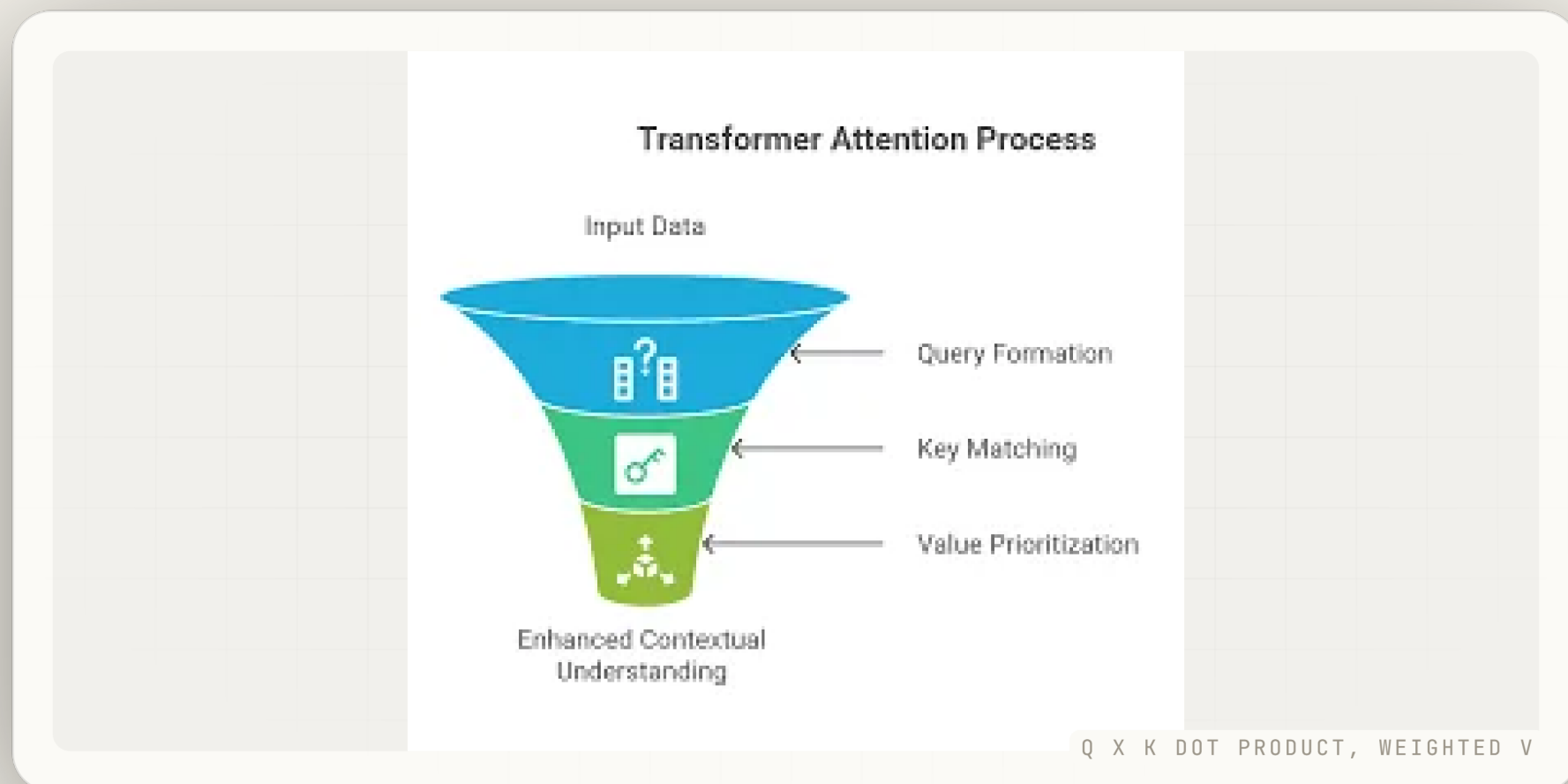
QKV

what to ask, what to match, what to return

QKV are the three vectors at the heart of attention: Query asks a question, Key signals relevance, Value carries the answer. **Every transformer ever built runs on this loop.**

USED IN

- Attention mechanism
- All transformers
- LLMs



QUALITATIVE VARIABLE • STATISTICS

Qualitative Variable

categories, not numbers

Qualitative variables represent labels and categories, not numbers. Color, gender, product type. **They require encoding before any ML model can use them** - a core data prep step.

USED IN

- Feature engineering
- Classification
- Data analysis

Qualitative vs. Quantitative



CATEGORICAL ENCODING APPROACHES

QUANTITATIVE VARIABLE • STATISTICS

Quantitative Variable

numbers you can do math on

Quantitative variables are measurable numbers: age, revenue, temperature. **They are the raw material of statistical analysis, regression, and nearly every ML model ever trained.**

USED IN

- Regression models
- Feature scaling
- Statistical analysis

Qualitative vs. Quantitative



CONTINUOUS VS DISCRETE NUMERIC

QUANTUM MACHINE LEARNING • EMERGING

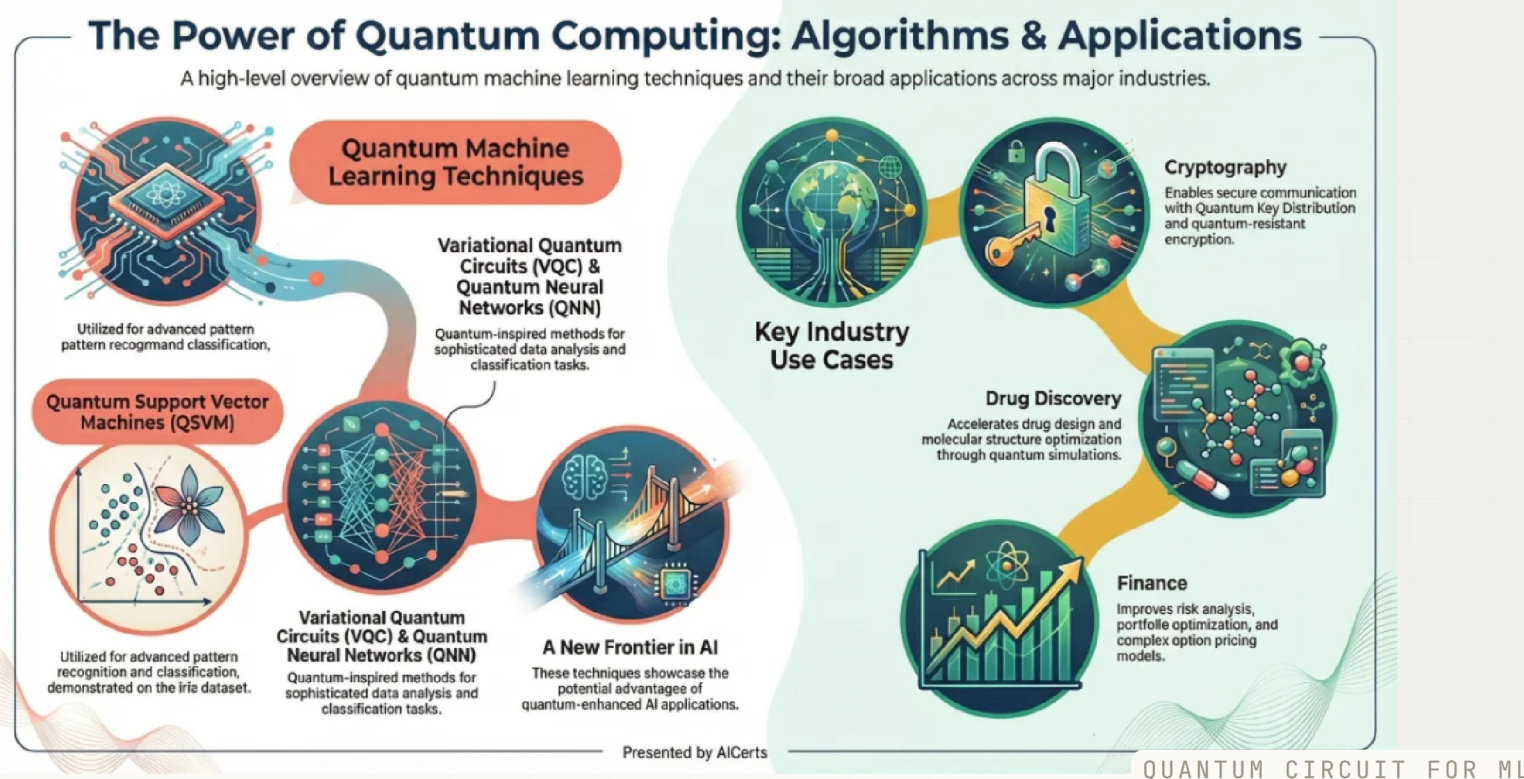
QML

quantum compute meets ML

QML applies quantum algorithms to ML problems, exploring whether **quantum superposition and entanglement can accelerate optimization and pattern recognition** beyond classical limits.

USED IN

- Quantum computing
- Research
- Optimization



Q-TABLE • REINFORCEMENT LEARNING

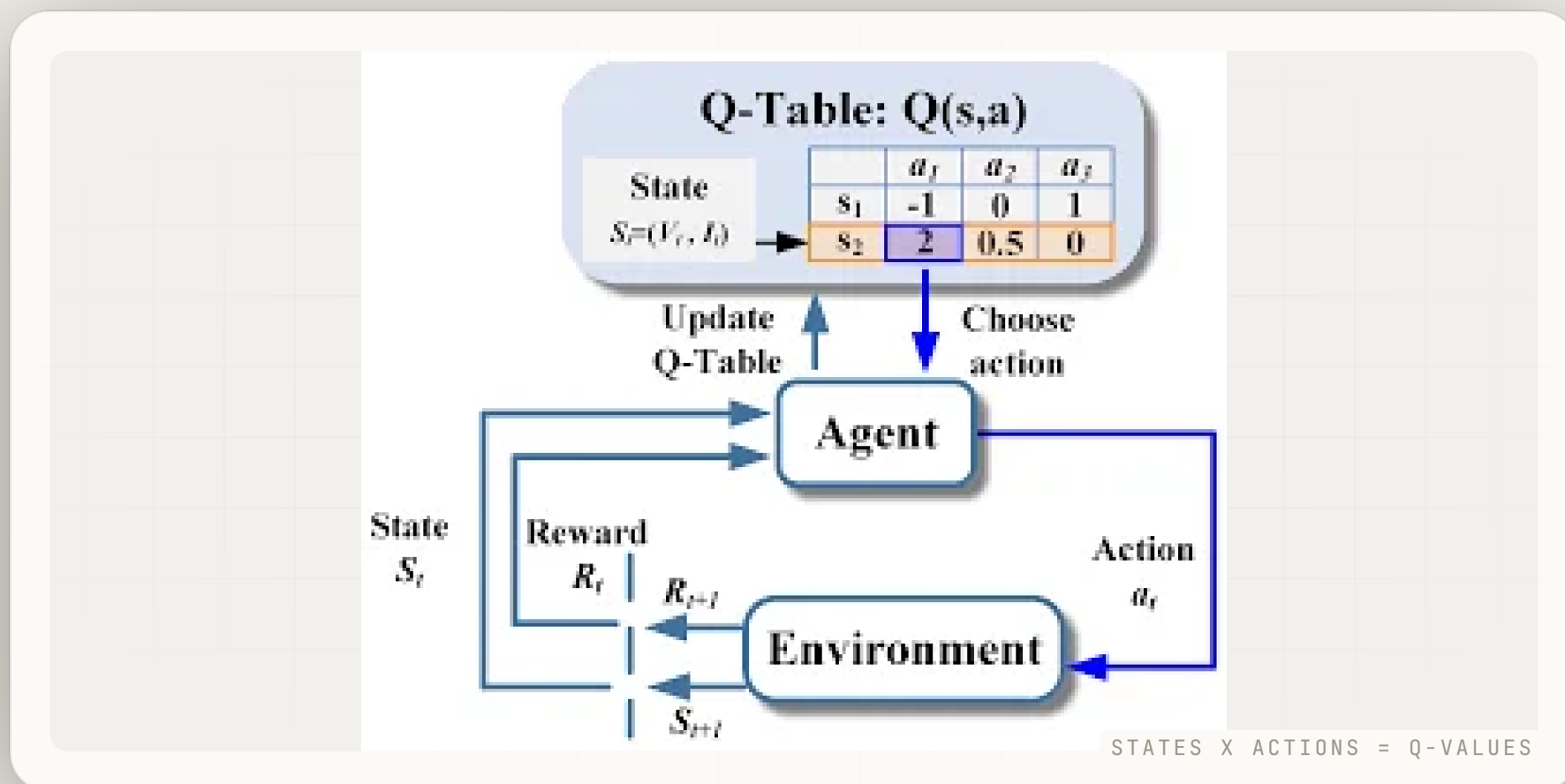
Q-Table

a map of every action's future reward

A Q-Table is the lookup table at the heart of Q-Learning. It stores the **expected future reward for every state-action pair**, letting the agent always pick the highest-value move.

USED IN

- Q-Learning
- Tabular RL
- Game AI



↓ UP NEXT • COMPANIES & TOOLS

The Q labs.

Three Q-named platforms: **open-weight models** from Alibaba, a vector database for RAG, and on-device AI from Qualcomm.

OPEN-WEIGHT MODEL FAMILY

Qwen

Alibaba Cloud • open weights • multimodal



Alibaba Cloud's open-weight LLM family covering **reasoning, coding, vision, and agentic tasks** - one of the most capable and widely adopted open-model ecosystems globally.

Alibaba

Developed by Alibaba Cloud

Open

Open-weight, HuggingFace releases

Multi

Text, code, vision, and agents

WHAT IT BUILDS

LLMs for reasoning, code, and vision.

FIND THEM

qwen.ai • huggingface.co/Qwen

WHY IT MATTERS

Top open-weight alternative to GPT-4.

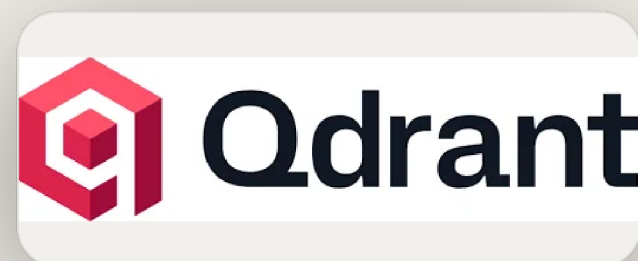
PARENT

Alibaba Cloud, China.

VECTOR DATABASE

Qdrant

open-source • RAG • semantic search



An open-source vector database built for **fast semantic search and RAG pipelines**, storing and querying embeddings at scale with filtering, payload storage, and cloud-native deployment.

Open

Open-source, Rust-based engine

RAG

Most popular use: retrieval pipelines

Fast

Designed for high-throughput search

WHAT IT DOES

Stores and queries vector embeddings.

BUILT WITH

Rust for speed and memory safety.

WHY IT MATTERS

Core infrastructure for RAG and search.

FIND THEM

qdrant.tech • github.com/qdrant

ON-DEVICE AI HARDWARE

Qualcomm AI



edge AI • NPU • Snapdragon

Qualcomm's AI ecosystem enables **on-device inference across phones, PCs, cars, and IoT** through dedicated NPUs, making AI fast and private without a cloud round-trip.

NPU

Dedicated neural processing units

Edge

AI running on the device, not cloud

Snapdragon

Inside billions of devices worldwide

WHAT IT ENABLES

AI inference on phones, PCs, and cars.

KEY PRODUCT

Snapdragon with Hexagon NPU.

WHY IT MATTERS

Privacy-first AI without cloud dependency.

FIND THEM

qualcomm.com/ai

↓ UP NEXT • CHANNEL TO FOLLOW

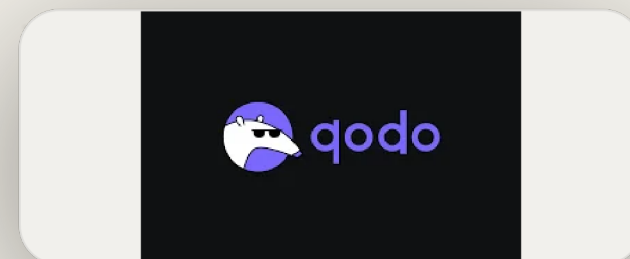
Learn from the best.

One Q-channel worth subscribing to: **AI-assisted coding** and developer tools explained clearly for engineers building real software.

AI DEV TOOLS CHANNEL

Qodo

@qodoai • code quality • AI dev



An AI-powered platform teaching **code quality, testing, and AI-assisted programming** through educational content and developer tools for modern software engineering workflows.

Testing

AI-generated test coverage

Review

AI-assisted code review

DevOps

Modern AI-native dev workflows

BEST FOR

Engineers adopting AI coding tools.

PLATFORM

YouTube + developer tooling at qodo.ai.

WHAT THEY COVER

Testing, review, and AI dev workflows.

FIND THEM

youtube.com/@qodoai • qodo.ai

↓ UP NEXT • PEOPLE TO FOLLOW

The Q minds.

Four Q-named researchers and educators **worth a follow** – from Google Brain founders to the person who made coding free for the world.

MULTI-AGENT AI RESEARCHER

Qingyun Wu

Microsoft AutoGen • @qingyun_wu



A core contributor to [Microsoft AutoGen](#), the multi-agent framework that lets AI agents collaborate, communicate, and solve complex tasks together autonomously.

AutoGen

Core contributor to MS AutoGen

Agents

Multi-agent collaboration research

Microsoft

Research at Microsoft

KNOWN FOR

AutoGen and multi-agent AI systems.

BEST FOR

Engineers building with AI agents.

WHY FOLLOW

Frontier agentic AI research in practice.

FIND THEM

x.com/qingyun_wu • qingyun-wu.github.io

AI RESEARCHER • GOOGLE BRAIN

Quoc Le

Google DeepMind • @quocleix



Co-founder of Google Brain and a pioneer in **large-scale neural networks, AutoML, and seq2seq learning** - his work shapes the foundation every modern LLM is built on.

Google Brain

Co-founder of the lab

Seq2Seq

Pioneered sequence-to-sequence models

AutoML

Neural architecture search research

KNOWN FOR

Google Brain, seq2seq, and AutoML.

NOW AT

Google DeepMind as VP of Research.

WHY FOLLOW

Foundational ideas behind modern LLMs.

FIND THEM

x.com/quocleix • research.google

AI RESEARCHER • FEDERATED LEARNING

Qiang Yang

HKUST • transfer learning • federated AI



The researcher who pioneered **transfer learning** and **federated learning** - showing how models learn across distributed data while keeping that data private.

Transfer

Pioneer of transfer learning theory

Federated

Privacy-preserving distributed ML

HKUST

Professor, Hong Kong

KNOWN FOR

Transfer learning and federated learning.

BEST FOR

Enterprise AI and data privacy research.

WHY FOLLOW

Privacy-first ML and cross-domain AI.

FIND THEM

cse.hkust.edu.hk/~qyang

EDUCATOR • FREECODECAMP

Quincy Larson

freeCodeCamp • @ossia



Founder of freeCodeCamp - the platform that has helped **millions of people learn to code for free**, including data science, ML, and AI, removing every barrier to tech education.

freeCodeCamp

Founded the platform in 2014

Free

Everything free, forever

Millions

Learners across 190+ countries

KNOWN FOR

freeCodeCamp and free tech education.

BEST FOR

Learners starting from zero.

WHY FOLLOW

Democratizing access to coding and AI.

FIND THEM

freecodecamp.org • linkedin.com/in/quincylarson

The Q stack, in one card.

Save it. Drop a comment with the Q-term you use most.

01 **Q-Learning**

RL

02 **Query Caching**

inference

03 **QLoRA**

fine-tuning

04 **Quantization**

optimization

05 **QKV**

transformers

06 **Qualitative Var.**

stats

07 **Quantitative Var.**

stats

08 **QML**

emerging

09 **Q-Table**

RL

+ **Qwen**

open models

+ **Qdrant • Qualcomm AI**

tools

► **Qodo**

channel

@ **Qingyun • Quoc • Qiang • Quincy**

people



FOLLOW FOR THE FULL A TO Z
Prasanna Kulkarni
in/prasanna-kulkarni-032650180

R Next